

**AMRITA VISHWA VIDYAPEETHAM
AMRITAPURI**

**B. Tech CSE (ARTIFICIAL
INTELLIGENCE)**

LITERATURE REVIEW SUBMISSION

Group No: 5

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1. Problem Definition

The research problem focuses on leveraging Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques to extract, analyze, and represent structured knowledge from Ayurvedic texts written in Sanskrit. Classical texts like the Charaka Saṃhitā and Suśruta Saṃhitā are rich sources of medical knowledge but pose challenges due to complex linguistic constructs such as sandhi, samāsa, free word order, and the morphological richness of Sanskrit. The goal is to design an AI-based framework that performs both syntactic and semantic analysis of these texts, leading to the construction of a queryable Knowledge Graph (KG). This KG will include entities such as DRAVYA (substances/herbs), ROGA (diseases), LAKSHANA (symptoms), and DOSHA (bioenergetic principles), along with their inter-relationships. The ultimate objective is to bridge traditional Ayurvedic wisdom with modern computational tools for knowledge retrieval, decision support, and further research in integrative medicine.

2. Literature Review Summary Table

No.	Author(s) & Proceedings	Title	Year	Dataset Used	Methodology / Approach	Limitations & Future Work
1	Maheshwari et al., Findings of EMNLP	A Benchmark and Dataset for Post-OCR text correction in Sanskrit	2022	218k Sanskrit OCR-corrected sentences	Seq2seq correction baselines for Sanskrit OCR	Needs Ayurveda-specific adaptation. not domain-focused
2	Zhu et al., Frontiers in Genetics (HerbKG)	HerbKG: A Knowledge Graph of Herbal Medicine	2022	PubMed abstracts, herbal texts	BERT-based NER & RE for Herb-Chemical-Gene-Disease KG	Focused on Chinese medicine; adaptation needed for Ayurveda
3	Nehrdich et al., Findings of EMNLP (ByT5-Sanskrit)	ByT5-Sanskrit: Multitask Pretraining for Sanskrit NLP	2024	Multitask Sanskrit corpora	Byte-level T5 pretrained on Sanskrit, multitask fine-tuning	No Ayurvedic domain adaptation yet

4	Tandra (Sujoy) et al., WSC Workshop	Building Named Entity Resources for Sanskrit Texts	2023	Śrīmad-Bhāgavatam subset (NER corpus)	Pre-annotation + bilingual alignment + gazetteers	Entities limited to persons/places. Ayurveda entities missing
5	Trappey et al., MDPI Information	Ontology-Based Knowledge Graph Framework for Text Mining	2024	Patent + industrial corpora	Intelligent text mining + ontology-based framework	Generic; not made for ancient Sanskrit text

3. Review of Selected Papers

Maheshwari et al. (2022) introduced a benchmark dataset of 218k OCR-corrected Sanskrit sentences, providing seq2seq correction baselines for post-OCR processing. While valuable for digitizing Sanskrit texts, it does not focus on Ayurveda-specific corpora.

Zhu et al. (2022) developed HerbKG, a knowledge graph of herbal medicine using PubMed abstracts and herbal texts. Their BERT-based NER and relation extraction capture herb–chemical–gene–disease interactions but are limited to Chinese medicine, requiring adaptation for Ayurveda.

Nehrdich et al. (2024) presented ByT5-Sanskrit, a byte-level multitask pretrained model for Sanskrit NLP. It demonstrates strong performance across diverse Sanskrit tasks but has not been fine-tuned for Ayurvedic texts.

Tandra et al. (2023) built a named entity recognition resource from a subset of the Śrīmad-Bhāgavatam using pre-annotation, bilingual alignment, and gazetteers. The corpus includes persons and places but lacks Ayurvedic entities such as herbs, diseases, and treatments.

Trappey et al. (2024) proposed an ontology-based framework for knowledge graph construction from patents and industrial corpora. Their approach is generic and ontology-driven but has not been applied to ancient Sanskrit medical texts.

4. Identified Research Gaps

- No domain-specific annotated Ayurvedic corpus exists.
- OCR correction methods exist but lack adaptation to Ayurvedic manuscripts.
- Knowledge graphs (HerbKG, ontology-based) are non-Sanskrit or non-Ayurvedic.
- NER resources exclude Ayurvedic entities.
- Pretrained Sanskrit PLMs (ByT5) not fine-tuned for Ayurveda.

5. Novelty / Contribution of Proposed Work

- .Creation of an Ayurveda-annotated corpus with herbs, diseases, treatments, and formulations.
- Fine-tuning ByT5-Sanskrit on Ayurvedic texts for NER and relation extraction.
- Semi-automatic KG construction using ontology alignment and NLP pipelines.
- Implementation of a Neo4j-based Ayurveda Knowledge Graph with query capability.
- Open-source release of datasets, models, and KG for research reproducibility.

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